AIGR TreeGrafter Function-Inference Stress Test — Final Report

Target: *Octopus vulgaris* DDO (UniProt **A0A7E6FSU6**, OXDD_OCTVU) **Seed hypothesis (TreeGrafter/PANTHER, IEA, GO_REF:0000118):** DDO has **D-amino acid catabolic process (GO:0019478) Focus:** function_assignment · `existing_annotations[5].function_hypothesis`

Summary

The TreeGrafter/PANTHER pipeline propagated the term **GO:0019478 (D-amino acid catabolic process)** to *Octopus vulgaris* DDO (A0A7E6FSU6). Independent, primary-evidence analysis **confirms this assignment is correct**. The query protein is a genuine, catalytically competent **D-aspartate oxidase (EC 1.4.3.1)** — a peroxisomal FAD-dependent flavoenzyme that oxidatively deaminates the acidic D-amino acids D-aspartate (and D-glutamate/NMDA) to the corresponding 2-oxo acid, ammonia, and hydrogen peroxide. This reaction is, by definition, a D-amino acid catabolic process, so the propagated biological-process term is accurate.

Critically, **none of the three characteristic automated-propagation failure modes apply.** (1) On *granularity*, the protein belongs to the correct subfamily — its closest characterized homolog is human D-aspartate oxidase (DDO, Q99489) at 41.4% identity, not the broad-specificity D-amino acid oxidase (DAAO) at ~36%. (2) On *pseudo-enzyme* status, the specific catalytic, substrate-selectivity, cofactor-binding, and targeting residues are all present and correctly spaced in both sequence and the AlphaFold model — the enzyme is active, not a degenerate fold. (3) On *within-superfamily mis-placement*, the sequence sits squarely in the acidic-D-amino-acid-oxidase clade, confirmed by the diagnostic acidic-substrate selectivity arginine R234.

The single most decisive piece of evidence is the intact, physically assembled active site: in the high-confidence AlphaFold model (mean pLDDT 95.3), the catalytic Arg275, selectivity Arg234, and active-site Tyr220 cluster within ~5–8 Å of one another — the geometric signature of a functional acidic-D-amino-acid oxidase — while octopus D-aspartate oxidase is an *experimentally established* enzyme historically used as a specific analytical reagent to quantify

D-aspartate (PMID: 8446003). The recommended curation action is to **retain** GO:0019478, optionally **making it more specific** by adding the molecular-function term **GO:0008445 (D-aspartate oxidase activity)**. This case is a clean **true-positive** for the TreeGrafter pipeline.

Executive Judgment

Verdict: SUPPORTED. The seed hypothesis holds. None of the three failure modes the stress test probes for are present:

Failure mode	Test applied	Result
(1) Granularity / wrong subfamily	Closest characterized homolog + diagnostic residue	Closest = human DDO (41.4%); selectivity R234 present → **correct subfamily**
(2) Pseudo-enzyme / lost activity	Catalytic/cofactor residue conservation & spacing (sequence + AlphaFold)	All residues intact, pocket assembled → **active enzyme**
(3) Within-superfamily mis-placement	DDO vs DAAO vs yeast panel; selectivity Arg	Sits in DDO clade; BP term correct either way → **not mis-placed**

Single most decisive piece of evidence: the intact, correctly-spaced catalytic cluster in the AlphaFold model (R275–R234 = 5.2 Å; R275–Y220 = 5.4 Å), combined with the experimental precedent that octopus D-aspartate oxidase is a real, active enzyme (PMID: 8446003).

Independent Family / Function Assignment

Independent of the propagated term, my own analysis assigns:

- **Specific molecular function: D-aspartate oxidase activity** (EC 1.4.3.1) — FAD-dependent oxidative deamination of acidic D-amino acids (D-aspartate, D-glutamate/NMDA).
- **Characterized homolog this is based on: human DDO (Q99489)**, the nearest characterized neighbor at 41.4% pairwise identity.

- **GO term I would assign:**
- Molecular function: **GO:0008445 (D-aspartate oxidase activity)** — *more specific* than, and the enzymatic basis for, the seed term.
- Biological process: **GO:0019478 (D-amino acid catabolic process)** — *same branch as the seed term*; correct as propagated.

Granularity relative to the seed term: the seed BP term is **correct (same branch)**; the optional MF addition (GO:0008445) is **more specific**. No sibling or different-branch reassignment is warranted.

Key Findings

Finding 1 — A0A7E6FSU6 is a bona-fide D-aspartate oxidase with intact catalytic machinery

UniProt curates A0A7E6FSU6 (OXDD_OCTVU) with the recommended name **D-aspartate oxidase, EC 1.4.3.1**. Its domain architecture is exactly what this enzyme requires: **Pfam PF01266 (DAO)**, **InterPro IPR023209 (DAO)** plus **IPR006181 (D-amino acid oxidase conserved site)** with a positive **PROSITE PS00677** match, and **PANTHER PTHR11530**. The entry also carries the UniProt keyword **"Direct protein sequencing"** — the protein was physically isolated and sequenced, strong evidence that this is a genuinely expressed gene product and not a pseudogene or prediction artifact.

A pairwise global alignment over the 336-residue protein places it closest to the **D-aspartate oxidase** subfamily rather than the generic DAAO subfamily:

Characterized homolog	Pairwise % identity
Human **DDO** (Q99489)	**41.4%** (top hit)
Human DAAO	36.3%
Pig DAAO	35.6%

The specific catalytic and substrate-recognition residues are conserved and correctly positioned (full table in *Active-Site / Placement Analysis* below). The catalytic arginine (human DDO R278 / pig DAAO R283) maps to octopus **R275**; the acidic-substrate selectivity arginine

(human DDO R237) maps to **R234**; the active-site tyrosine (pig DAAO Y228) maps to **Y220**; the N-terminal FAD Rossmann fingerprint (GxGxxG) is present as **G8-AGVVGL**; and the C-terminal peroxisomal targeting signal (PTS1) is present as **SKL**. The one non-conserved basic position (human DDO R216 → octopus **Y213**) is distal to the active site and does not affect catalysis. The principal selectivity arginine (R234) — the residue that distinguishes DDO from DAAO by anchoring the side-chain carboxylate of D-aspartate — is fully intact.

Finding 2 — GO:0019478 is correctly propagated; it could be made more specific

D-aspartate oxidase (EC 1.4.3.1) oxidatively deaminates acidic D-amino acids (D-Asp, D-Glu/NMDA) to the 2-oxo acid + NH_3 + H_2O_2 — by definition a **D-amino acid catabolic process (GO:0019478)**. The octopus is an *established natural source* of this enzyme: **"Octopus D-aspartate oxidase" is used as a specific analytical reagent to quantify D-aspartate** (PMID: 8446003), and free D-Asp occurs specifically in cephalopod (*Octopus vulgaris*, *Loligo*, *Sepia*) nervous tissue. There is thus independent biochemical precedent that the octopus genome encodes a catalytically active D-aspartate oxidase, and the sequence/active-site analysis confirms A0A7E6FSU6 is that enzyme. The matching molecular-function term **GO:0008445 (D-aspartate oxidase activity)** is the specific activity underlying the propagated BP term.

> *Verbatim support (PMID: 8446003):* "D-Asp was discovered using specific methods that employ **Octopus D-aspartate oxidase** and hog kidney D-amino acid oxidase to measure D-Asp levels."

Finding 3 — AlphaFold geometry confirms an assembled, intact active site

The AlphaFold DB model **AF-A0A7E6FSU6-F1 (v6, mean pLDDT 95.3)** is high-confidence. The three key residues cluster tightly in three dimensions, forming a compact substrate-binding/catalytic pocket:

Residue pair	Distance	pLDDT
Catalytic R275 – selectivity R234	5.2 Å	97 / 94
Catalytic R275 – active-site Y220	5.4 Å	97 / 97
Selectivity R234 – active-site Y220	7.6 Å	94 / 97
Non-conserved Y213 – R275	21.5 Å	distal

Two arginines positioned ~5 Å apart is exactly the arrangement expected for anchoring **both** carboxylate groups of D-aspartate — the structural signature of the acidic-amino-acid oxidase (DDO), not the broad-specificity DAAO. The single non-conserved position (Y213, equivalent to human DDO R216) sits **16–21 Å from the catalytic cluster**, i.e. outside the active site, so its substitution is mechanistically irrelevant. The C-terminal **PTS1 (…S334-K335-L336)** is on a solvent-exposed, lower-pLDDT (82–88) terminus — the appropriate context for Pex5 recognition and peroxisomal import. The N-terminal FAD Rossmann motif centroid lies ~17 Å from the catalytic arginine, consistent with canonical DAO-fold geometry.

Finding 4 — Subfamily placement across a 9-homolog panel; BP term robust to DDO/DAAO ambiguity

Benchmarking against nine characterized homologs reinforces the DDO assignment:

Homolog	% identity to octopus
Human **DDO**	**41.4%** (best overall)
Rat DDO	37.8%
Mouse DAAO	37.8%
Mouse DDO	36.6%
Human DAAO	36.3%
Rat DAAO	36.3%
Pig DAAO	35.6%
Bovine DDO	32.4%
Yeast RgDAAO	23.8%
Mean DDOs	**37.1%**
Mean DAAOs	**33.9%**

The single best match is human D-aspartate oxidase, and the mean identity to DDOs (37.1%) exceeds that to DAAOs (33.9%). At invertebrate-to-vertebrate evolutionary distance, global identity alone does not cleanly separate DDO from DAAO (the best DAAO, mouse, reaches 37.8%). The *decisive* subfamily feature is therefore the conserved **acidic-substrate selectivity arginine R234** (= human DDO R237), reinforced by UniProt's curated naming and the literature reagent use of octopus D-aspartate oxidase. Crucially, **the propagated BP term (GO:0019478) is robust to the residual DDO/DAAO ambiguity**: both DDO and DAAO catabolize D-amino acids, so the seed term is correct either way; the distinction only affects the optional, more-specific MF term (GO:0008445 vs GO:0003884).

Mechanistic Model / Interpretation

The enzyme is a peroxisomal FAD-dependent oxidase operating on acidic D-amino acids:

```
D-aspartate  +  O2  +  H2O
       │
       ▼   [ FAD-dependent oxidative deamination ]
oxaloacetate  +  NH3  +  H2O2
```

Structural logic of the active site (AlphaFold AF-A0A7E6FSU6-F1):

```
     FAD (Rossmann, G8-AGVVGL)
           │  ~17 Å
           ▼
┌──────────────────────────────────┐
│   D-aspartate binding pocket     │
│                                  │
│   R275 (catalytic)               │
│     │5.2 Å        │5.4 Å       │
│   R234 (selectivity)  Y220       │
│        └──7.6 Å──┘               │
└──────────────────────────────────┘
           ▲
Y213 (non-conserved, 16–21 Å away — outside pocket, irrelevant)

C-terminus: …S-K-L (PTS1) → Pex5 → peroxisome
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The two arginines ~5 Å apart form a dual-carboxylate clamp that selects for **acidic** D-amino acids (D-Asp, D-Glu/NMDA), distinguishing this DDO subfamily from the broad-specificity DAAO subfamily, which lacks the second selectivity arginine. The FAD cofactor accepts the hydride during oxidative deamination and is re-oxidized by O_2, generating H_2O_2. The PTS1 (SKL) tripeptide targets the enzyme to the peroxisome, the canonical compartment for this reaction. This is internally consistent biochemistry end to end: a correct fold, a correctly assembled and correctly specific active site, the correct cofactor-binding motif, and the correct subcellular targeting signal all point to a single coherent function.

Biologically, this fits the known neurochemistry of cephalopods, where free D-aspartate accumulates in nervous tissue and a dedicated D-aspartate oxidase provides the catabolic counterpart. The net conclusion is that TreeGrafter got this one right; the case serves as a valuable **true-positive control** for the AIGR stress test — a propagated IEA term that withstands independent primary-evidence scrutiny.

Evidence Base

#	Citation	Evidence type	Call	Claim tested	Key finding	Context
1	UniProt A0A7E6FSU6 (OXDD_OCTVU)	review/ database	**Supports**	Identity/ annotation	RecName D-aspartate oxidase, EC 1.4.3.1; PF01266, IPR023209+IPR006181 (PS00677 match), PTHR11530; keyword "Direct protein sequencing"	*O. vulgaris*
2	Global alignment (this work)	computational	**Supports**	Subfamily placement	41.4% id to human DDO vs ~36% to DAAO → closest neighbor is D-aspartate oxidase	in silico
3	Residue mapping (this work)	structural/ evolutionary	**Supports**	Active site intact	Catalytic Arg→R275; selectivity Arg→R234; active Tyr→Y220; FAD GxGxxG; PTS1 SKL all conserved	in silico
4	PMID: 8446003 — *Occurrence of free D-aspartic acid in the circumsoesophageal ganglia of Aplysia fasciata*	direct assay / reagent use	**Supports**	Enzyme is real & active	"Octopus D-aspartate oxidase" used as a specific reagent to quantify D-Asp	mollusc neurochem
5	PMID: 24237903 — *Degradation of human DAAO and pLG72*	review	**Qualifies**	Family biochemistry/ compartment	hDAAO (EC 1.4.3.3) is a peroxisomal flavoenzyme in D-serine catabolism	human bra U87

#	Citation	Evidence type	Call	Claim tested	Key finding	Context
6	AlphaFold AF-A0A7E6FSU6-F1 v6	structural/ computational	**Supports**	Active site assembled	R275/R234/Y220 within 5–8 Å (pLDDT 94–97); Y213 16–21 Å away; PTS1 SKL exposed	in silico (pl 95.3)

How the two literature papers bear on the findings. PMID: 8446003 is the key positive anchor: it establishes that an *active* octopus D-aspartate oxidase exists and is reliable enough to be used as a specific analytical reagent — directly corroborating that the octopus DDO gene product carries genuine catalytic activity. PMID: 24237903 describes the sibling enzyme human DAAO as a peroxisomal flavoenzyme central to brain D-serine catabolism; it supports the family-level mechanism and compartment while underscoring the DDO-vs-DAAO distinction that the active-site analysis resolves.

Active-Site / Placement Analysis (computed)

Catalytic residues: INTACT. Pairwise global alignment of A0A7E6FSU6 against characterized members, with residues mapped from reference numbering to octopus positions:

Reference position	Role	Octopus residue	Conserved?	AlphaFold position
human DDO R278 / pig DAAO R283	catalytic Arg (binds α-carboxylate)	**R275**	Yes	Core of pocket (pLDDT 97)
human DDO R237	acidic-substrate selectivity Arg (DDO diagnostic)	**R234**	Yes	5.2 Å from R275 (pLDDT 94)
pig DAAO Y228	active-site Tyr	**Y220**	Yes	5.4 Å from R275 (pLDDT 97)
N-term FAD Rossmann GxGxxG	FAD binding	**G8-AGVVGL**	Yes	~17 Å from catalytic Arg
human DDO R216	second DDO basic residue	**Y213**	No (Arg→Tyr)	16–21 Å from pocket (distal, irrelevant)
C-term PTS1	peroxisomal targeting	**S334-K335-L336 (SKL)**	Yes	Solvent-exposed C-terminus (pLDDT 82–88)

Interpretation. Identity: octopus vs human DDO **41.4%**, vs human DAAO 36.3%, vs pig DAAO 35.6% (~331–333 aligned columns each). **Catalytic residues are intact → not a pseudo-enzyme; subfamily is D-aspartate oxidase → not mis-placed.** The structural confirmation from AlphaFold (mean pLDDT 95.3) shows the catalytic/substrate residues clustering into a compact, high-confidence active site (R275–R234 = 5.2 Å, R275–Y220 = 5.4 Å, R234–Y220 = 7.6 Å, all pLDDT 94–97), confirming the pocket is physically assembled. The single non-conserved position (Y213 = human DDO R216) lies 16–21 Å from the catalytic cluster, so the substitution flagged in the sequence analysis is **structurally peripheral and functionally irrelevant**. The sequence therefore belongs to the **acidic-D-amino-acid-oxidase (DDO) subfamily** of the DAO flavoprotein fold, with the conserved selectivity arginine R234 as the decisive discriminator.

GO Curation Implications

Recommended curation action (lead requiring curator verification):

- **GO:0019478 (D-amino acid catabolic process), BP — RETAIN.** The propagated IEA term is correct and well-supported by independent primary evidence. The IEA/GO_REF:0000118 propagation is accurate here and can be upgraded with the sequence- and structure-based evidence above.
- **Optionally MAKE-MORE-SPECIFIC** by adding the molecular-function term **GO:0008445 (D-aspartate oxidase activity)** / EC 1.4.3.1, which the sequence and active-site evidence directly support.
- Do **not** remove, generalize, replace with a sibling term, treat as non-core, or fall back to "protein binding." The annotation is core and accurate.

This is a **true-positive** TreeGrafter propagation and a useful positive control for the AIGR pipeline.

Limitations and Knowledge Gaps

1. **No direct enzymatic assay on this exact accession.** Catalytic competence is inferred from sequence conservation, AlphaFold geometry, and the general precedent that octopus D-aspartate oxidase is an active, reagent-grade enzyme (PMID: 8446003). A kinetic assay on recombinant A0A7E6FSU6 would convert high-confidence inference into direct evidence.
2. **DDO vs DAAO not cleanly separable by global identity.** At invertebrate-to-vertebrate distance, the best DAAO homolog (mouse, 37.8%) is close to the DDO mean (37.1%). The DDO call rests principally on the single diagnostic selectivity arginine (R234) plus curated naming — a narrow structural feature. Residual DAAO-like activity on neutral/basic D-amino acids cannot be excluded without substrate-profiling. Importantly, this ambiguity does **not** affect the BP term under test.
3. **AlphaFold is a model, not an experimental structure.** The pocket geometry (R275–R234 = 5.2 Å, etc.) is a high-confidence prediction (pLDDT 95.3), but no crystal or cryo-EM structure of this protein exists to confirm it.
4. **Accession-to-literature link is indirect.** The historical reagent "octopus D-aspartate oxidase" predates this genome-derived accession; their identity is highly plausible but not formally proven.

5. **GO_REF:0000118 / IEA provenance** was taken as given; the reference-tree graft itself was not re-run here.

Proposed Follow-up Experiments / Actions

1. **Recombinant kinetic assay (most decisive).** Express A0A7E6FSU6 in *E. coli* or yeast and measure activity on D-aspartate and D-glutamate/NMDA (expected DDO substrates) versus neutral/basic D-amino acids (D-serine, D-alanine, D-proline — DAAO substrates). A high D-Asp/D-Ser activity ratio confirms strict DDO identity and would justify locking in GO:0008445.
2. **Substrate-selectivity mutagenesis.** Mutate R234→Y (mimicking the DAAO-like loss of the second selectivity arginine) and test whether acidic-substrate preference is abolished — a direct functional validation that R234 is the DDO determinant.
3. **Peroxisomal localization check.** Confirm the PTS1 (SKL) functions by expressing a GFP-fusion in a cephalopod or heterologous cell line and verifying peroxisomal import, closing the loop on the compartment.
4. **Experimental structure.** A crystal structure with a bound substrate/analog (e.g., D-aspartate or a malate/tartrate mimic) would verify the AlphaFold-predicted dual-arginine clamp.
5. **Curatorial action.** Flag the annotation as a verified true-positive and, pending or in lieu of assay, add GO:0008445 as an ISS/IEA molecular-function term with the active-site evidence as the justification.

Report generated from a 3-iteration autonomous investigation. Findings F001–F004 and literature PMIDs 8446003 and 24237903 as recorded in the knowledge state.
